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INTRODUCTION TO INDUSTRIAL ENGINEERING

Lecture #13 Work Design

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Agenda of the Day

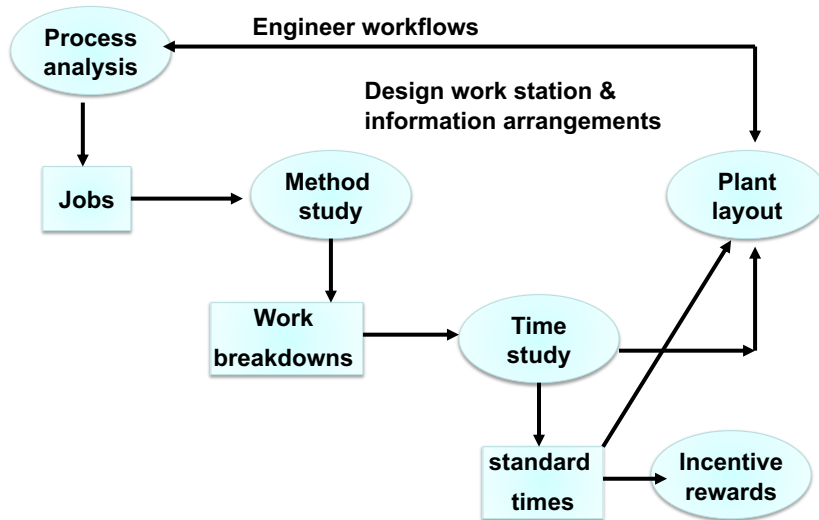
- Work Design - Job Design
- Method Study
- Time Study

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An overview of interactions



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Job Design

- ❑ Who does what, how and where?
- ❑ Should be consistent with organizational goals.
- ❑ Everybody in the system should understand these and hence should be communicated properly.
- ❑ Involvement of employees and top management is important.

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Method Study

- ❑ Method (motion) study is the principal technique for reducing the work involved, primarily by eliminating unnecessary movement on the part of material or operatives and by substituting good methods for poor ones.
- ❑ Flow diagrams and process charts are the tools.

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Comparison of Method and Time Study

- ❑ Method Study (Gilbreth)
 - Improve methods
 - It measures distance, or how much you move to do a job, and how much you get done in a period of time
- ❑ Time study (Taylor)
 - Establish standards
 - Looked at the average time it took an average worker

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Work simplification

- ❑ Work Simplification -- process of changing work methods:
 - Eliminate unnecessary parts of work (non-value-added steps)
 - Combine and rearrange parts of work
 - Simplify work when possible
- ❑ Work Simplification Tools
 - Flow Process Chart (can eliminate, combine, change (sequence, place, person), or improve activities)
 - Layout Chart

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Advantages and Disadvantages of Specialization

	Management	Employees
Advantages	a. Simplifies training b. Higher productivity c. Low wage costs	a. Low education/skill b. Minimum responsibilities c. Little mental effort needed
Disadvantages	a. Difficult to motivate quality b. Worker dissatisfaction, absenteeism, high turn-over, disruptive tactics, poor attention to quality	a. Monotonous and boring b. Limited opportunities for advancement c. Little control over work d. Little opportunity for self-fulfillment

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Does efficiency sufficient for productivity?

- ❑ Satisfaction of wants and needs is important when humans are involved.
- ❑ This means we need intrinsic and extrinsic motivators to minimize the boredom caused by monotony of specialization.
- ❑ We have to include human focus (behavioral side).

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Job improvement

- ❑ *Job enlargement* -- give workers a larger portion of the total task (horizontal expansion -- additional work at same level of skill and responsibility)
 - In a manufacturing setting, job enlargement might mean having a worker do several tasks at a work station rather than only one or two.
 - In a bank, it might mean training a person to write car loans, and installment loans rather than only one of these.

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Job improvement

- ▣ *Job enrichment* -- increasing responsibility for planning and coordinating tasks (vertical expansion)
 - It may mean that a production worker is involved in the design of the product or production process is responsible for his own quality testing, handles customer complaints, or deals directly with suppliers.

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Job improvement

- ▣ *Job rotation* -- workers periodically exchange jobs
 - Cross-training is necessary to shift workers to other tasks.
 - Broadens understanding and can reduce fatigue.

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Work Measurement

- **Work measurement** is the process of establishing the time that a given task would take when performed by a qualified worker working at a defined level of performance.
- **A qualified worker** is one who has acquired the skill, knowledge and other attributes to carry out the work in hand to satisfactory standards of quantity, quality and safety.
- **Work measurement** also refers to the process of estimating the amount of worker time required to produce one unit of output.

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Work Measurement

- Work measurement helps to uncover non-standardization that exist in the workplace and non-value adding activities and waste.

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When to use?

- ❑ To provide information on which the **planning and scheduling of production** can be based, including the plant and labor requirements for carrying out the program of work and the utilization of available capacity.
- ❑ To provide information on which **estimates for tenders, selling prices and delivery promises** can be based.
- ❑ To set standards of **machine utilization and labor performance** which can be used for any of the above purposes and as a basis **for incentive schemes**.
- ❑ To provide information for labor-cost control and to enable **standard costs** to be fixed and maintained.

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Why Work Measurement?

- ❑ A work has to be measured for the following reasons:
 - To discover and eliminate lost or ineffective time.
 - To establish standard times for performance measurement.
 - To measure performance against realistic expectations.
 - **To set operating goals and objectives.**

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Techniques for WM

- ❑ **Two alternative method:**
 - **Direct work measurement:**
 - **Time study:** “the most widely used”
 - **Work sampling:** Determines the proportion of time a worker spends on activities
 - **Indirect work measurement** (indirect method not on the task) with standard times are measured by standard data and formulation.
 - Synthesis from standard data.
 - Estimating.
 - Analytical estimating.
 - Comparative estimating.

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Work Measurement

- ❑ Main method is Time Study, which is the development of a standard time by observing a task and analyzing it with the use of a stopwatch.
- ❑ Time Study includes:
 - Establishment of standard times - management knowledge
 - Rating operator performance - criteria for appraisal
 - Gathering information to calculate production capabilities and data for capacity planning.
 - Defining work content of finished goods and services e.g. for charging and estimating.

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Standard Time

- ❑ The amount of time it should take a qualified worker to complete a specified task, working at a sustainable rate, using given methods and equipment, raw materials, and workplace arrangements is called a **standard time**.
- ❑ A **Standard time** can be developed through:
 - Stop-watch studies (time study)
 - Historical times
 - Predetermined data
 - Work sampling

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Method for Time Study

- ❑ Take time over a number of trials
 - ❑ Workers should be educated regarding the process to avoid suspicion
 - ❑ Number of trials to time (i.e., cycles or sample size) depends on:
 - variability in observed times
 - desired accuracy
 - desired level of confidence for the estimate
- Requires knowledge of probability concepts:

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Determining «Standard Time»

- ▣ *Observed Time* -- average of observed times
 - $OT = \sum x_i / n$
- ▣ *Normal Time* -- observed time adjusted for worker performance
 - The time required for an average, trained operator to perform a task under usual working conditions and working at a normal pace. (It does not include allowances for personal needs and delays that would be necessary if the task were done all 8 hours.)

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Determining «Standard Time»

- ▣ *Observed Time* -- average of observed times
 - $OT = \sum x_i / n$
- ▣ *Normal Time* -- observed time adjusted for worker performance
 - $NT = OT * PR$ (where PR = performance rating measured for the entire job)
 - $NT = \sum (E_j * PR_j)$ (where PR is measured element by element)
 - PR equals 1 for the average worker; $PR < 1$ is for a slower worker

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$$ST = NT * (1+AF)$$

- ❑ **Standard time** equals normal time multiplied by an allowance factor
- ❑ **Allowance Factor:** The amount of time for personal needs, unavoidable delays, and fatigue. It should be given as a percentage. If the allowances are given in terms of time units, must be converted into percentages.

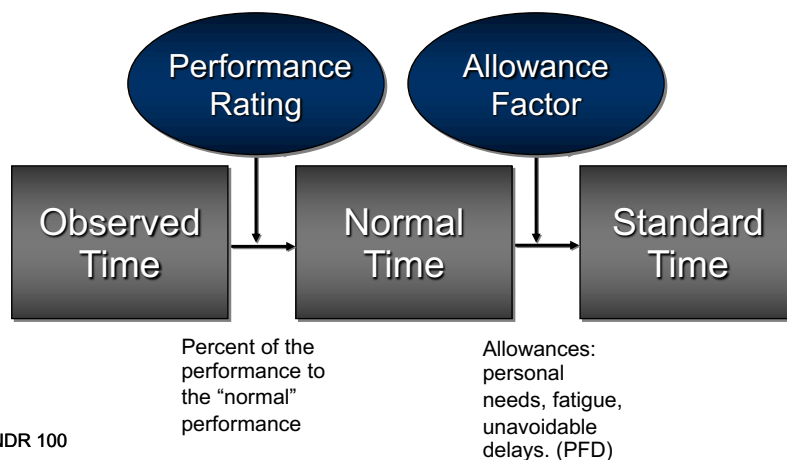
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Determining Labor Standards from Time Studies

$$\text{Standard time} = OT * PR * (1 + AF)$$



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Typical Allowance Percentages for Varying Healthcare Delivery Working Conditions

Allowance Level	Percent
1. Basic-low (personal, fatigue, standing)	11
2. Basic-moderate (basic-low and mental strain)	12
3. Basic-high (basic-moderate and slightly uncomfortable heat/cold or humidity)	14
4. Medium-low (basic high and awkward position)	16
5. Medium-moderate (medium-low and lifting requirements up to 20 lbs.)	19
6. Medium-high (medium-moderate and loud noise)	21
7. Extensive-low (medium-high and tedious nature of work)	23
8. Extensive-medium (extensive-low and with complex mental strain)	26
9. Extensive-high (extensive-medium and lifting requirement up to 30 lbs.)	28

Source: Adapted from B.W. Niebel, 1988.

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Example 1: Time Study Approach

In a time study of a manufacturing operation, 10 actual times are recorded for the duration of producing a product:

The average time observed to complete a product was 8.6 minutes. The performance rating applied to the observed worker was 0.95 and the allowance during an 8-hour shift was 12.5% or 60 minutes.

9.28	9.20
8.57	7.73
8.04	7.87
9.25	9.73
7.70	8.98

Compute the standard time for this work.



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Example 1: Time Study Approach

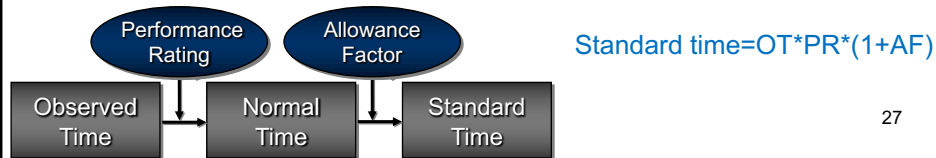
Observed time = 8.6 minutes

Performance rating = 0.95

Allowance factor = 0.125

Normal time = Observed time x Performance rating
 $= 8.6 \times 0.95$
 $= 8.17$ minutes

Standard Time = Normal time * (1 + Allowance factor)
 $= 8.17 * (1 + 0.125)$
 $= 8.17 * 1.125$
 $= 9.191$ minutes



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Class Exercise

- An assembly task has an **established normal time of 4.0 minutes per unit** based on previous studies with well-trained workers.
 - A time-study analyst now observes a new worker performing **10 cycles** of the same task and records the following **observed times (minutes)**.
 - The company uses an **allowance factor of 15%** of normal time for rest, personal needs, and unavoidable delays.
 - What is the standard time?
- | Cycle | Time |
|-------|------|
| 1 | 3.8 |
| 2 | 4.1 |
| 3 | 3.9 |
| 4 | 12.0 |
| 5 | 3.7 |
| 6 | 3.8 |
| 7 | 4.2 |
| 8 | 3.9 |
| 9 | 4.0 |
| 10 | 3.8 |

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Extra Credit Challenge

- A new worker is performing a repetitive assembly task. The task time decreases with experience and approaches a **minimum achievable time** as learning occurs.
- To model this effect, they use the following **asymptotic exponential learning curve**:

$$T(n) = x + c e^{-\alpha n}$$

where:

- $T(n)$ is the expected time of the n -th repetition,
- x is long-run minimum (limit) time as $n \rightarrow \infty$,
- $c = 1.5$ is initial learning gap,
- $\alpha = -0.3$ is learning rate constant.
- Performance rating of the worker is 0.9 and the company uses an allowance factor of %15.
- Consider the following measurements. What is the standard time?

n	1	2	3
Observed time	4.55	4.25	4.10

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Solution

- **Write the total error**

$$\sum_{n=1}^3 (y_n - [x + 1.5e^{-0.3n}])^2.$$

- **Differentiate and find x^* .**

$$\frac{dS}{dx} = -2 \sum_{n=1}^3 (y_n - x - 1.5e^{-0.3n}) = 0$$

$$\Rightarrow \sum_{n=1}^3 (y_n - x - 1.5e^{-0.3n}) = 0$$

$$\Rightarrow 3x = \sum_{n=1}^3 (y_n - 1.5e^{-0.3n})$$

$$x^* = \frac{1}{3} \sum_{n=1}^3 (y_n - 1.5e^{-0.3n})$$

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Solution

□ For $n = 1$

$$- y_1 - 1.5e^{-0.3} = 4.55 - 1.5(0.740818) \\ = 4.55 - 1.111227 = 3.438773$$

□ For $n = 2$

$$- y_2 - 1.5e^{-0.6} = 4.25 - 1.5(0.548812) \\ = 4.25 - 0.823218 = 3.426782$$

□ For $n = 3$

$$- y_3 - 1.5e^{-0.9} = 4.10 - 1.5(0.406570) \\ = 4.10 - 0.609855 = 3.490145$$

□ Now average:

$$x^* = \frac{3.438773 + 3.426782 + 3.490145}{3} = 3.45$$

$$\text{Standard Time} = 3.45 \times 0.9 \times 1.15 = 3.57$$

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Example 2: If a job has different elements

Job Element	Cycle Observed (in minutes)					PR
	1	2	3	4	5	
Compose and type letter	8	10	9	21*	11	90%
Type envelope address	2	3	2	1	3	85%
Stuff, stamp, and seal.	2	1	5*	2	1	94%

Allowance Factor =15%

- Delete unusual or nonrecurring observations (marked with *)
21 in job A, 5 in Job C

- Compute average times for each element

Average observed time of job A = $(8+10+9+11)/4 = 9.5 \text{ min.}$

Average observed time of job B = $(2+3+2+1+3)/5 = 2.2 \text{ min.}$

Average observed time of job C = $(2+1+2+1)/4 = 1.5 \text{ min.}$

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Example 2 (cont.)

- ❑ Compute the normal time for each element
- ❑ **NT = Average observed time x PR**
 - NT of job A = $9.5 \times 0.9 = 8.55\text{min}$
 - NT of job B = $2.2 \times 0.85 = 1.87 \text{ min}$
 - NT of job C = $1.5 \times 0.94 = 1.41 \text{ min}$
- ➔ Total Normal Time = $8.55 + 1.87 + 1.41 = 11.83 \text{ min}$
- ➔ Standard Time = $\text{NT} * (1 + \text{AF})$
- ➔ ST = $11.83 (1 + 0.15) = 13.6 \text{ min}$

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What are the problems with time studies?

- ❑ Subjective performance ratings and allowances
- ❑ Only observable jobs can be studied
- ❑ Highly costly -- best for repetitive tasks
- ❑ Disrupts worker routine
- ❑ May cause worker resentment

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